



S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

PRE-LAB

Aim: An Introduction to Python Programming Lab:

- To install and configure the Python Interpreter and explore various Python development environments such as Google Colab, IDLE, Visual Studio Code (VS Code), Jupyter Notebook, and PyCharm for writing, executing, and testing Python programs.

Name of Student:

Roll No.:

Semester/Year:

Academic Session:

Date of Performance:

Date of Submission:

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OBJECTIVES:

- Understand the importance of Python programming and its applications.
- Install and configure the Python Interpreter on a computer.
- Explore different Python development environments and their features.
- Write, execute, edit, and debug simple Python programs.
- Compare the advantages and limitations of different Python IDEs and online platforms..

Theory:

Python is a high-level, interpreted, object-oriented programming language developed by **Guido van Rossum** and first released in **1991**. It is widely used in software development, web development, artificial intelligence, machine learning, data science, automation, cybersecurity, and scientific computing because of its simple syntax and extensive library support.

Unlike compiled languages, Python executes code line by line using an interpreter, making debugging easier and reducing development time. Python is platform-independent, meaning programs written on one operating system can run on others with little or no modification.

Before writing Python programs, a suitable programming environment must be installed. Several development environments are available, each designed for different purposes.

1. Python Interpreter

The Python Interpreter converts Python source code into executable instructions. It provides an interactive shell where users can write and execute Python statements one at a time.

Features:

- Executes Python code line by line.
- Helps beginners learn programming interactively.
- Useful for testing small code snippets.

2. IDLE (Integrated Development and Learning Environment)

IDLE is the default IDE that comes with the official Python installation.

Features:

- Simple and lightweight.
- Built-in syntax highlighting.
- Auto indentation.
- Interactive Python Shell.
- Suitable for beginners.

3. Google Colab

Google Colaboratory (Colab) is a cloud-based Jupyter Notebook environment provided by Google.

Features:

- No installation required.
- Runs directly in a web browser.

- Free access to GPUs and TPUs.
- Easy sharing using Google Drive.
- Ideal for Data Science and Machine Learning.

Advantages

- Accessible from anywhere.
- Automatic saving.
- Easy collaboration.

4. Visual Studio Code (VS Code)

VS Code is a lightweight yet powerful source code editor developed by Microsoft.

Features

- Python extension support.
- Intelligent code completion (IntelliSense).
- Integrated terminal.
- Debugging support.
- Git integration.

Advantages

- Fast and customizable.
- Suitable for professional software development.

5. Jupyter Notebook

Jupyter Notebook is an interactive environment that combines executable code, text, images, and graphs in one document.

Features

- Interactive coding.
- Supports Markdown documentation.
- Visualization using charts and graphs.
- Ideal for Data Analysis and Research.

6. PyCharm

PyCharm is a professional Python IDE developed by JetBrains.

Features

- Smart code completion.
- Powerful debugging tools.
- Project management.
- Version control integration.
- Virtual environment support.

Advantages

- Excellent for large Python projects.
- Preferred by professional developers.

Environment	Installation Required	Best For	Key Feature
Python Interpreter	Yes	Learning Basics	Interactive execution
IDLE	Yes	Beginners	Simple interface
Google Colab	No	AI & Machine Learning	Cloud-based with GPU support
VS Code	Yes	Software Development	Extensions and debugging
Jupyter Notebook	Yes	Data Science	Interactive notebooks
PyCharm	Yes	Professional Development	Advanced coding features

Advantages of Python

- Easy to learn and understand.
- Free and open source.
- Platform independent.
- Large standard library.
- Supports multiple programming paradigms.
- Strong community support.
- Widely used in industry.

Applications of Python

- Web Development
- Machine Learning
- Artificial Intelligence
- Data Science
- Automation
- Cybersecurity
- Game Development
- Desktop Applications
- Internet of Things (IoT)
- Cloud Computing

Installation Procedure with Expected Outputs (IMPLEMENTATION / SCREENSHOTS)

DISCUSSION AND VIVA VOCE:

Q. What is Python?

Ans.

Q. What is the purpose of the Python Interpreter?

Ans.

Q. Which Python environment is best suited for Machine Learning?

Ans.

Q. What is the difference between IDLE and VS Code?

Ans.

Q. Why is Python considered platform independent?

Ans.

CONCLUSION:

This practical introduced the Python programming environment and familiarized students with installing and configuring the Python Interpreter. Students explored popular development environments such as **IDLE, Google Colab, VS Code, Jupyter Notebook, and PyCharm**, understanding their features, advantages, and suitable applications. By completing this practical, students gained the foundational knowledge required to write, execute, and test Python programs efficiently using different development tools.